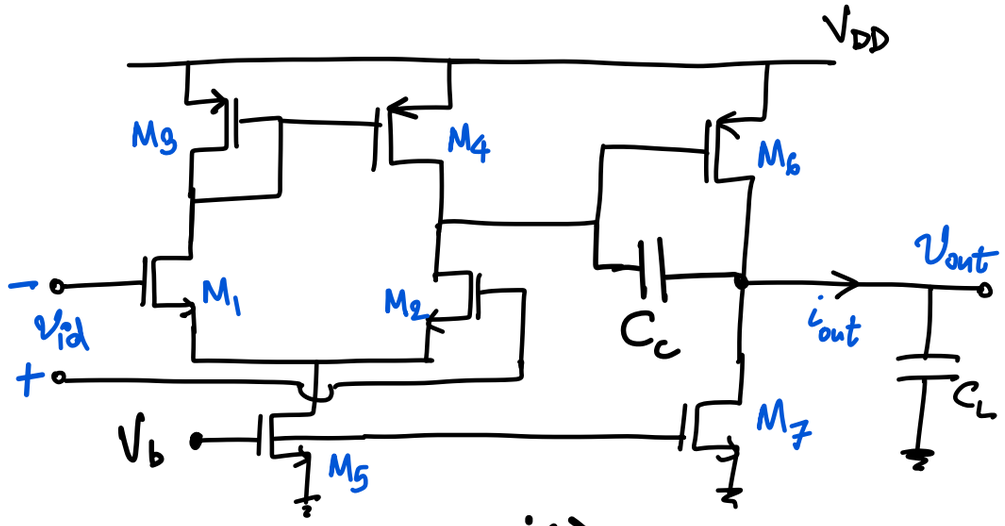
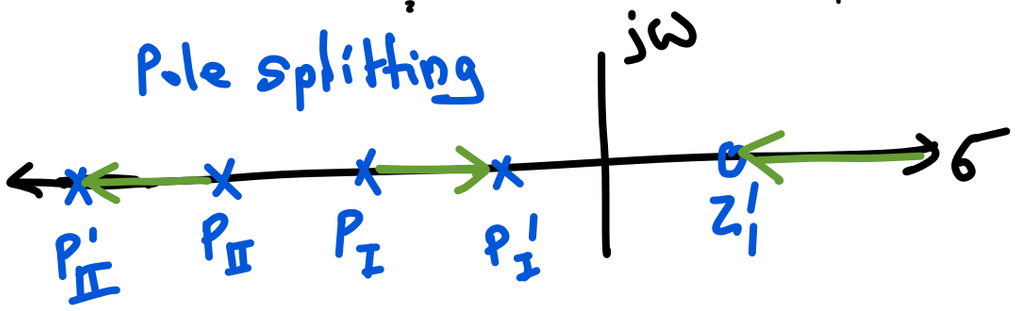
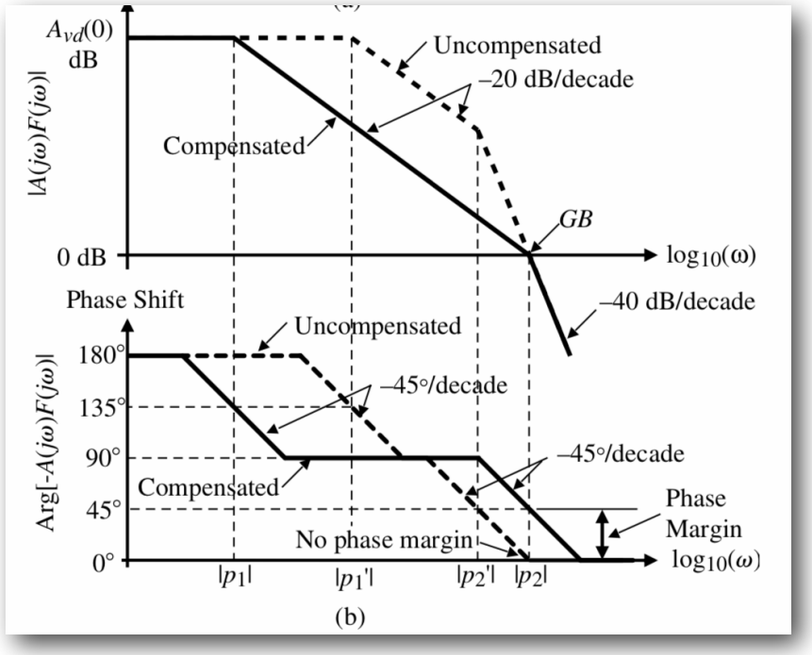
Pole splitting (an important concept)







$P_I = \frac{-1}{R_I C_I}$

$$P_I' = \frac{-1}{R_I \cdot g_{m_{II}} R_{II} \cdot C_c}$$

(Pole moves towards origin)

$P_{II} = \frac{-1}{R_{II} \cdot C_{II}}$

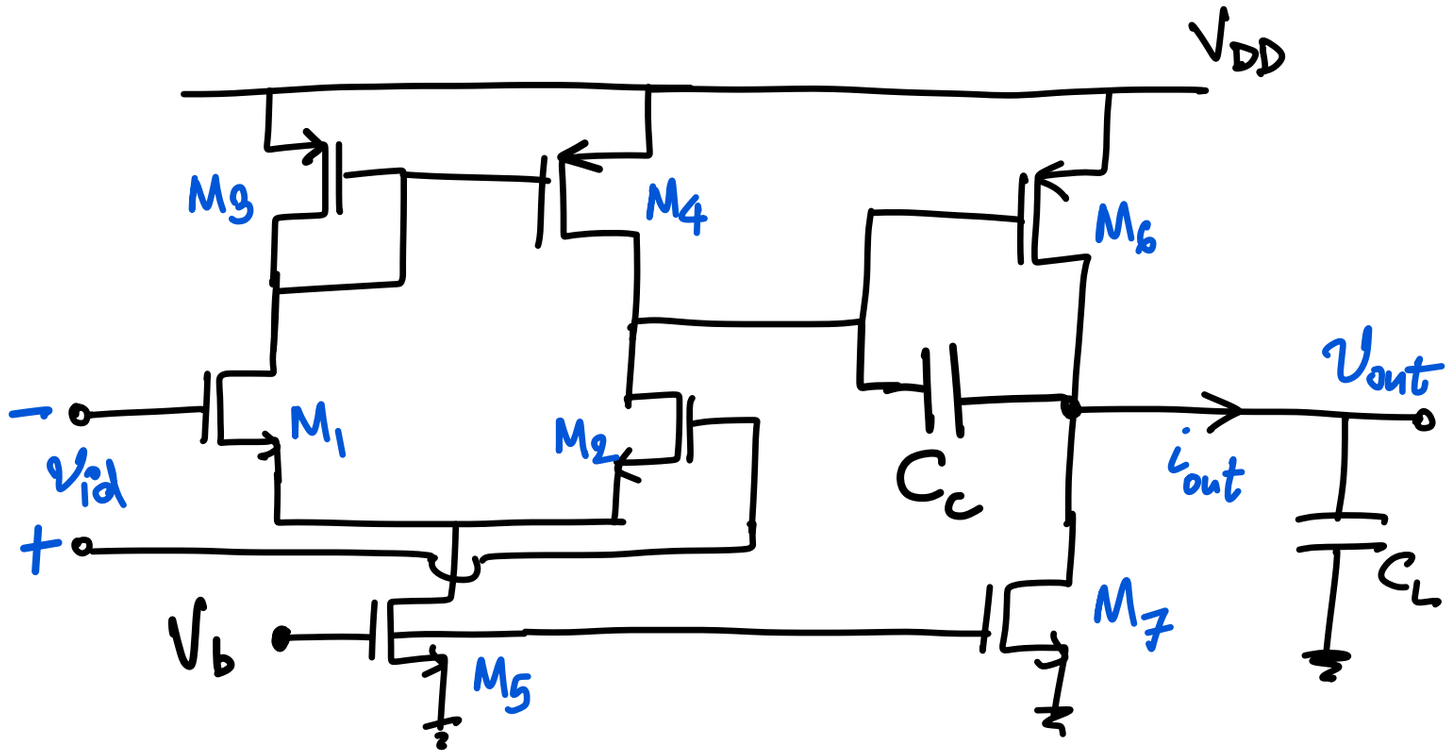
$$P_{II}' = -\frac{g_{m_{II}}}{C_{II}} \approx \frac{g_{m6}}{C_L}$$

(Pole moves away from origin)

$$\left[g_{m_{II}} R_{II} \gg 1 \Rightarrow g_{m_{II}} \gg \frac{1}{R_{II}} \right]$$

$Z_1' = \frac{g_{m_{II}}}{C_c}$

[No zero without C_c. $\Rightarrow Z_1 \to \infty$]



$$A_{v0} = g_{m_I} g_{m_{II}} \cdot R_I \cdot R_{II}$$

$$p_1' = -\frac{1}{R_I \cdot g_{m_{II}} R_{II} \cdot C_c}$$

$$p_2' = -\frac{g_{m_{II}}}{C_{II}}$$

$$z_1' = \frac{g_{m_{II}}}{C_c} \quad (?)$$

$$GBW = |A_{v0}| \cdot |p_1|$$

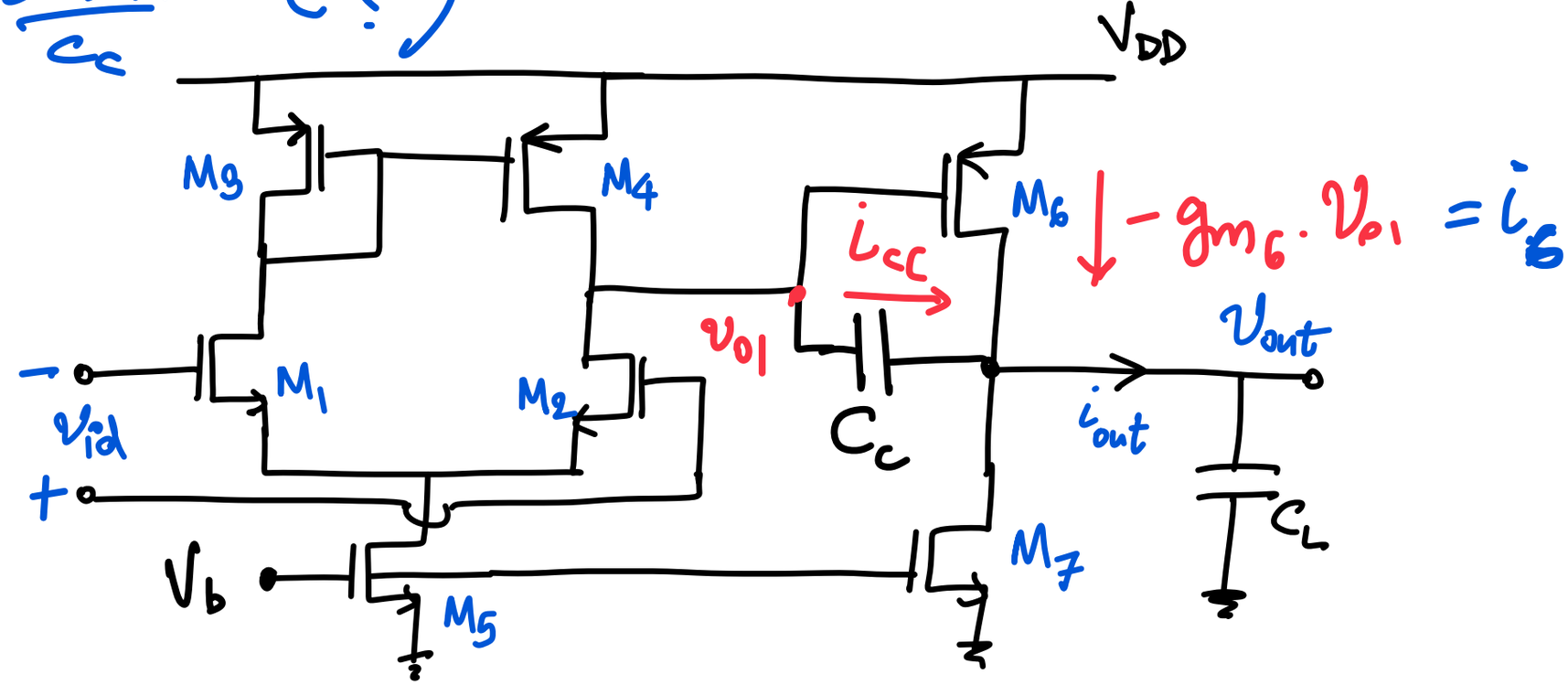
$$= \frac{g_{m_I}}{C_c}$$

where, $g_{m_I} = g_{m_1} = g_{m_2}$

$g_{m_{II}} = g_{m_6}$

Explanation of the origin of RHP zero.

$z_1' = \frac{g_{m1}}{C_c}$ (?)

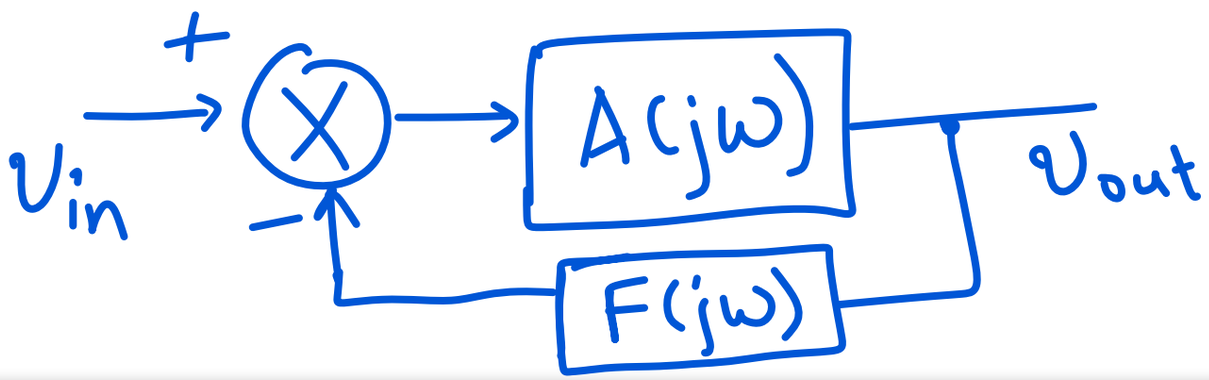


$i_{cc} = (v_{o1} - v_{out}) C_c . s$

At a particular freq of v_{o1}, $i_{cc} + i_6 = 0 \Rightarrow v_{out} = 0$

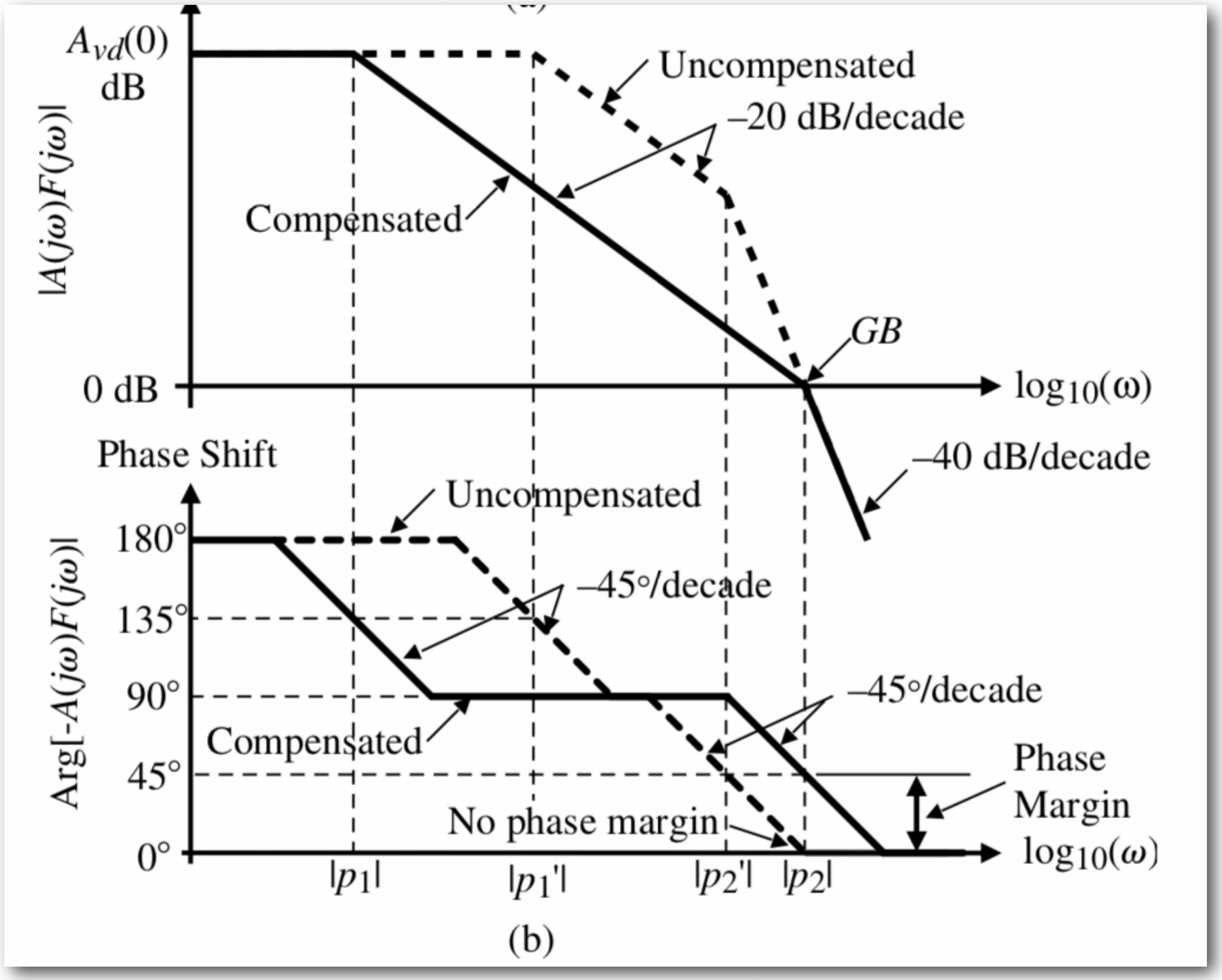
$\Rightarrow (v_{o1} - v_{out}) C_c s = g_{m6} . v_{o1} \Rightarrow v_{o1} - v_{out} = \frac{g_{m6}}{C_c s} . v_{o1}$

$\Rightarrow v_{out} = v_{o1}\left(1 - \frac{g_{m6}}{C_c s}\right) = 0 \Rightarrow \frac{g_{m6}}{C_c s} = 1 \Rightarrow z' = \frac{g_{m6}}{C_c}$



$$\frac{v_{out}}{v_{in}} = \frac{A(j\omega)}{1 + A(j\omega)F(j\omega)}$$

$$L(j\omega) = -A(j\omega)F(j\omega)$$



Example:

For an OTA, with two poles and one zero, prove that if the zero is ten times higher than GB, then in order to achieve phase margin $\phi_m = 60°$, the second pole must be placed at least 2.2 times higher than GB.

Soln:

$$A_v(s) = \frac{-A_{v0} \cdot \left(\frac{s}{z_1} - 1\right)}{\left(\frac{s}{p_1} + 1\right)\left(\frac{s}{p_2} + 1\right)}$$

$$\phi_M = 180° - \tan^{-1}\left(\frac{\omega}{p_1}\right) - \tan^{-1}\left(\frac{\omega}{p_2}\right) - \tan^{-1}\left(\frac{\omega}{z_1}\right) = 60°$$

Example:

$$\phi_M = 180^\circ - \tan^{-1}\left(\frac{\omega}{p_1}\right) - \tan^{-1}\left(\frac{\omega}{p_2}\right) - \tan^{-1}\left(\frac{\omega}{z_1}\right) = 60^\circ$$

Assuming the unity gain frequency is GB,

$$120^\circ = \tan^{-1}\left(\frac{GB}{p_1}\right) + \tan^{-1}\left(\frac{GB}{p_2}\right) + \tan^{-1}\left(\frac{GB}{z_1}\right)$$

$$\Rightarrow 120^\circ = \tan^{-1}(A_{v0}) + \tan^{-1}\left(\frac{GB}{p_2}\right) + \tan^{-1}(0.1)$$

$$\Rightarrow 120^\circ - 90^\circ - 5.7^\circ \approx \tan^{-1}\left(\frac{GB}{p_2}\right)$$

$$\Rightarrow 24.3^\circ \approx \tan^{-1}\left(\frac{GB}{p_2}\right) \Rightarrow p_2 \approx 2.2\,GB$$

Example:

$p_2 \approx 2.2\,GB$

Assuming that a min. of 60° ϕ_m is req., the following relationships apply:

$$\frac{g_{m6}}{C_c} > 10\left(\frac{g_{m2}}{C_c}\right) \Rightarrow \boxed{g_{m6} > 10 g_{m2}}$$

(Zero: g_{m6}, UGB: g_{m2})

$$\frac{g_{m6}}{C_{II}} > 2.2\left(\frac{g_{m2}}{C_c}\right) \Rightarrow \boxed{C_c > 0.22\, C_{II}}$$

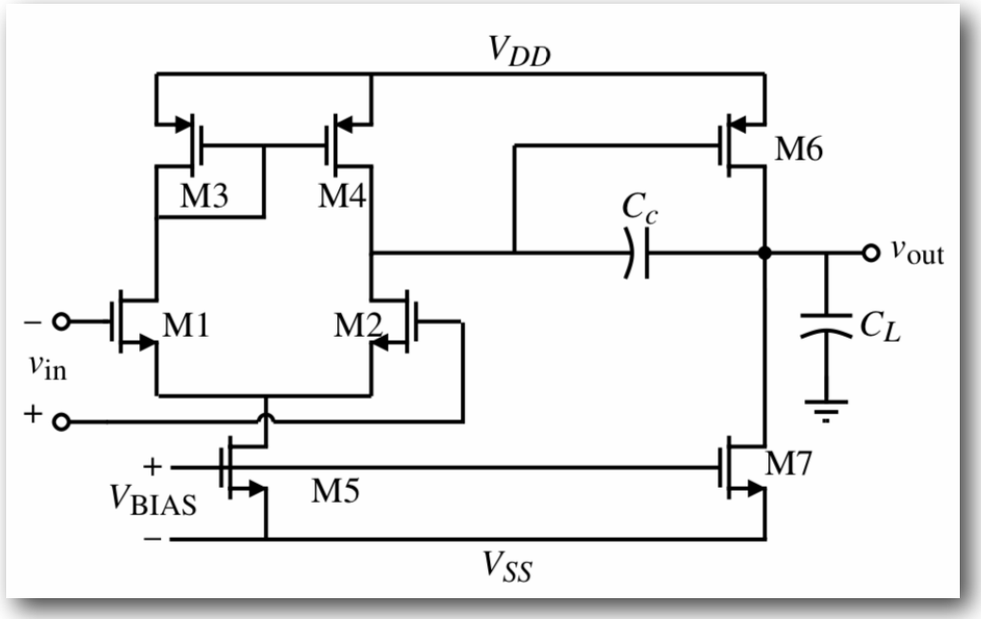
(p_2: g_{m6}/C_{II}, UGB: g_{m2}/C_c)

First stage gain, $A_{v1} = -g_{m1} \cdot (r_{o2} \| r_{o4})$

$$= -\frac{g_{m1}}{\frac{I_5}{2}(\lambda_2 + \lambda_4)}$$

Second stage gain, $A_{v2} = -g_{m6} \cdot (r_{o6} \| r_{o7})$

$$= -\frac{g_{m6}}{I_6(\lambda_6 + \lambda_7)}$$



$$GB = \frac{g_{m1}}{C_c}, \quad p_2 = -\frac{g_{m6}}{C_L}, \quad z_1 = \frac{g_{m6}}{C_c}$$

$$V_{in(max)} = ?$$

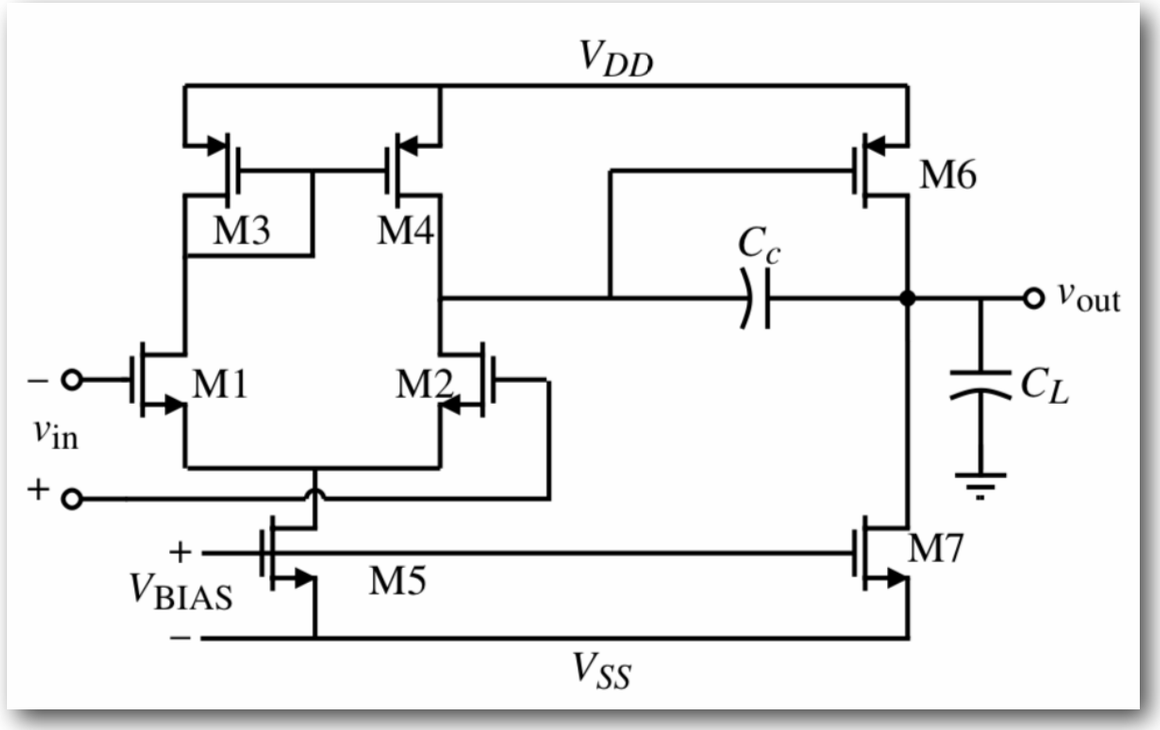
$$I_3 = \frac{1}{2} \cdot \beta_3 \cdot (V_{SG3} - |V_{THP3}|)^2$$

$$\sqrt{\frac{2I_3}{\beta_3}} = V_{SG3} - |V_{THP3}|$$

$$\sqrt{\frac{I_5}{\beta_3}} = V_{DD} - V_{G3} - |V_{THP3}|$$

$$\sqrt{\frac{I_5}{\beta_3}} = V_{DD} - [V_{in(max)} - V_{THN1}] - |V_{THP3}|$$

$$\Rightarrow V_{in(max)} = V_{DD} - \sqrt{\frac{I_5}{\beta_3}} + V_{THN1} - |V_{THP3}|$$

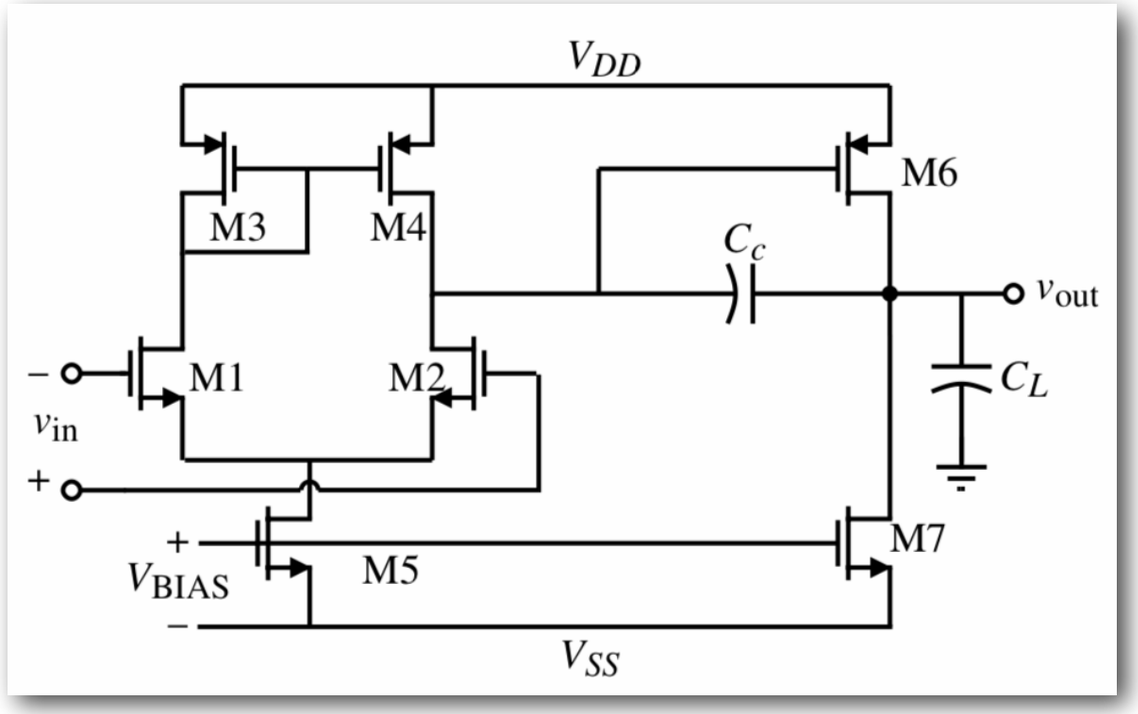


$V_{in(min)} = ?$

$$I_1 = \frac{1}{2} \cdot \beta_1 \cdot (V_{GS} - V_{THN1})^2$$

$$\sqrt{\frac{I_5}{\beta_1}} = V_{in(min)} - V_{S1} - V_{THN1}$$

$$= V_{in(min)} - [V_{SS} + V_{DSS(sat)}] - V_{THN1}$$

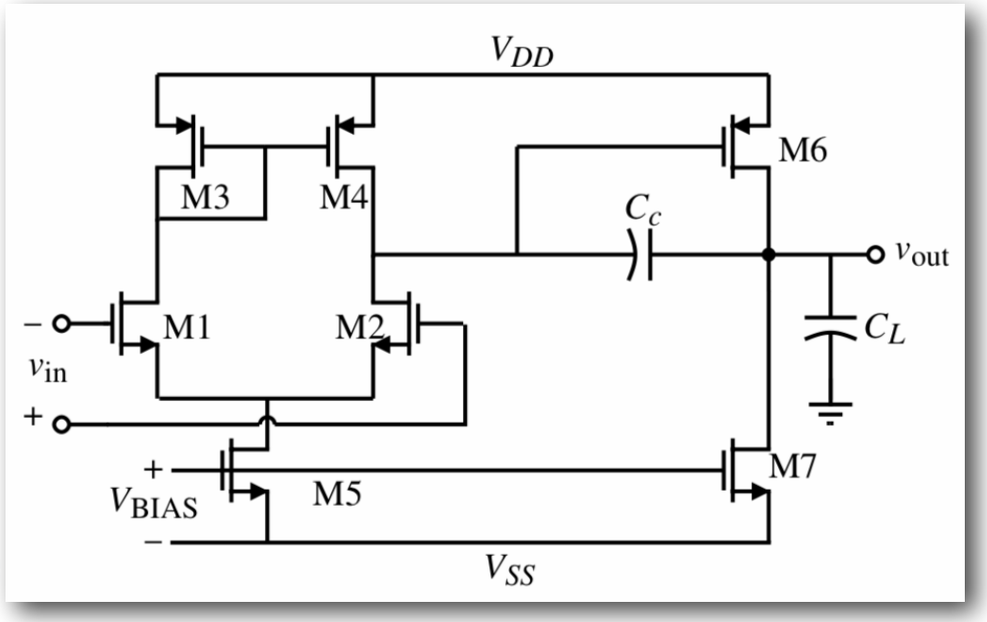


$$V_{in(min)} = \sqrt{\frac{I_5}{\beta_1}} + V_{THN1} + V_{DSS(sat)} + V_{SS}$$

Saturation voltage $V_{DS(sat)} = \sqrt{\frac{2 I_{DS}}{\beta}}$
(or over drive (V_{OV}))

First stage gain, $A_{v1} = -g_{m1} \cdot (r_{o2} \| r_{o4})$

$$= \frac{-2 g_{m1}}{I_5 (\lambda_2 + \lambda_4)}$$



Second stage gain, $A_{v2} = -g_{m6} \cdot (r_{o6} \| r_{o7})$

$$= -\frac{g_{m6}}{I_6 (\lambda_6 + \lambda_7)}$$

$$GB = \frac{g_{m1}}{C_c}, \quad p_2 = -\frac{g_{m6}}{C_L}, \quad z_1 = \frac{g_{m6}}{C_c}$$

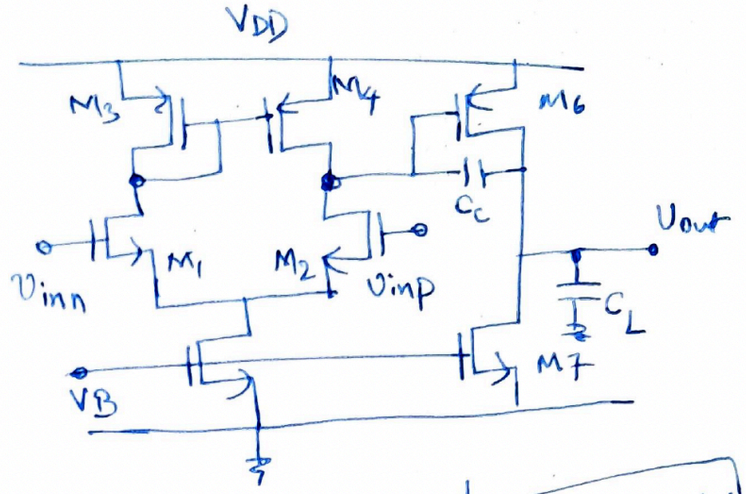
$$V_{in(max)} = V_{DD} - \sqrt{\frac{I_5}{\beta_3}} + V_{THN1} - |V_{THP3}|$$

$$V_{in(min)} = V_{SS} + \sqrt{\frac{I_5}{\beta_1}} + V_{THN1} + V_{DS5(sat)}$$

Slew rate, $SR = I_5 / C_c$ (V/s)

① $L = 1\,\mu m$

Component	Sizing
M_1	✓
M_2	✓
M_3	✓
M_4	✓
M_5	✓
M_6	✓
M_7	✓
I_{D5}	✓
I_{D6}	✓
C_C	✓



I

C_L → C_C : $\because \phi_m = 60^\circ$, $C_C > 0.22 C_L$

→ I_5 : $\because I_5 = SR \times C_C$

→ I_3 : $\because I_3 = \frac{I_5}{2}$

→ $\left(\frac{W}{L}\right)_{3,4}$

$$= \frac{2 I_3}{K'_{P3}\left[V_{DD} - \left(V_{in,max} - V_{THN}\right) - |V_{THP}|\right]^2}$$

Check:

$$\frac{g_{m3,4}}{2 C_{gs3}} > 10\ GB$$

Calculate

$$g_{m3,4} = \sqrt{2 \mu_p C_{ox} \left(\frac{W}{L}\right)_{3,4} \cdot I_3}$$

II

$$\frac{g_{m1}}{C_C} = GB$$

$$g_{m1} = C_C \times GB$$

→ $\left(\frac{W}{L}\right)_{1,2}$

$$= \frac{g_{m1,2}^2}{2 \mu_n C_{ox} I_3}$$

→ $\left(\frac{W}{L}\right)_5$

$$= \frac{2 I_5}{K'_{n5} \cdot [V_{ov5}]^2}$$

where

~~$V_{ov5} = V_{in(min)}$~~

~~$- V_{TH1}$~~

~~$- V_{ov}$~~

$V_{ov5} = V_{in(min)} - V_{ov1} - V_{TH1}$

III

$$\frac{g_{m6}}{C_L} > 2.2\ GB$$

$$\frac{g_{m6}}{C_L} > 2.2 \frac{g_{m1}}{C_C}$$

$$\Rightarrow g_{m6} = 2.2\, g_{m1} \left(\frac{C_L}{C_C}\right)$$

$V_{SG4} = V_{SG6}$

$$\left(\frac{W}{L}\right)_6 = \frac{g_{m6}}{g_{m4}} \cdot \left(\frac{W}{L}\right)_4$$

$(\because g_m = k'_p \left(\frac{W}{L}\right) V_{ov})$

I_{D6}

$$= \frac{g_{m6}^2}{2 \mu_p C_{ox} \left(\frac{W}{L}\right)_6}$$

$I_{D7} = I_{D6}$

$$\left(\frac{W}{L}\right)_7 = \left(\frac{W}{L}\right)_5 \cdot \frac{I_{D6}}{I_{D5}}$$

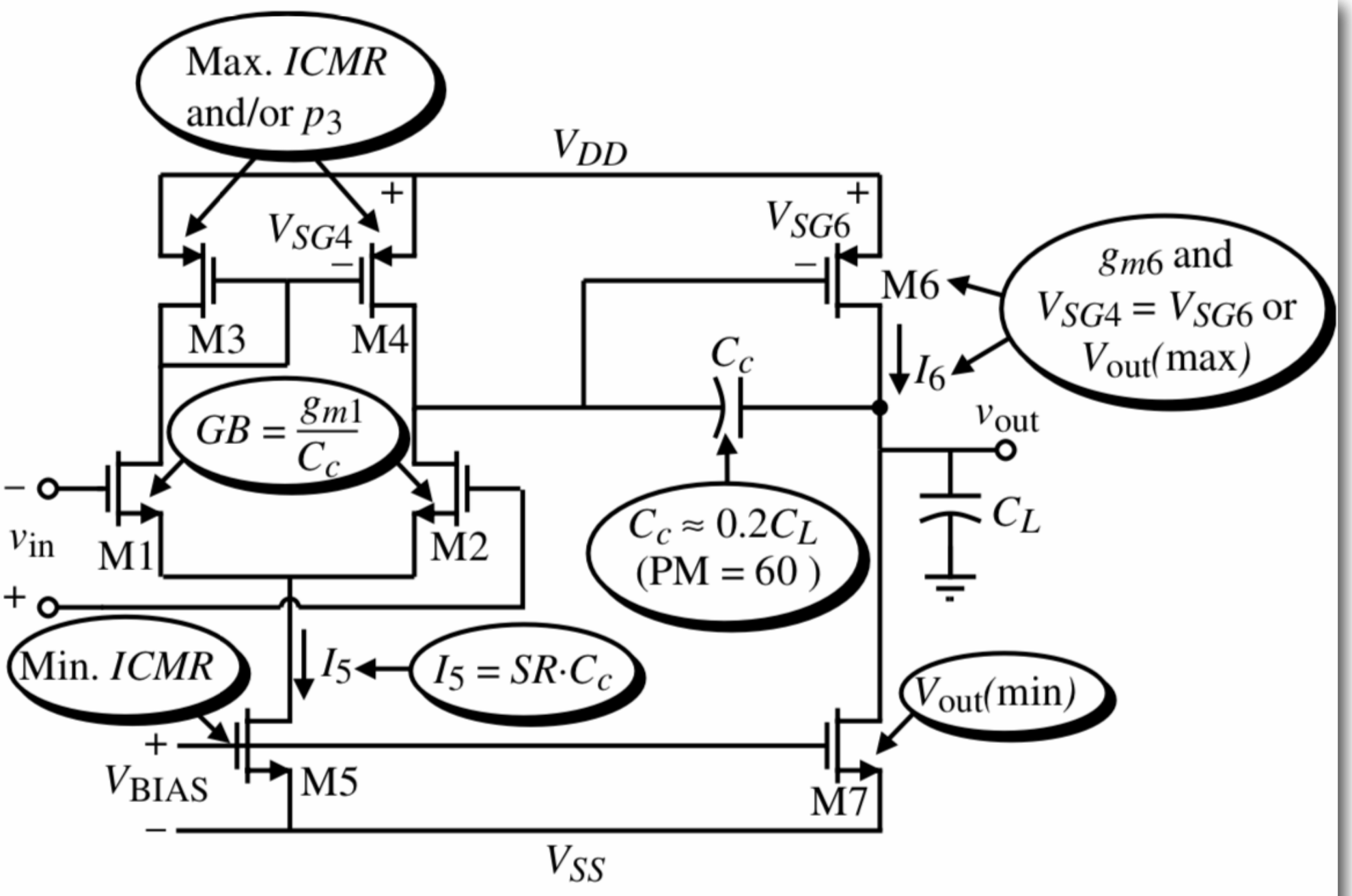


Figure 6.3-2 Illustration of the design relationship and the circuit for a two-stage CMOS op amp.

Simulation test-benches:

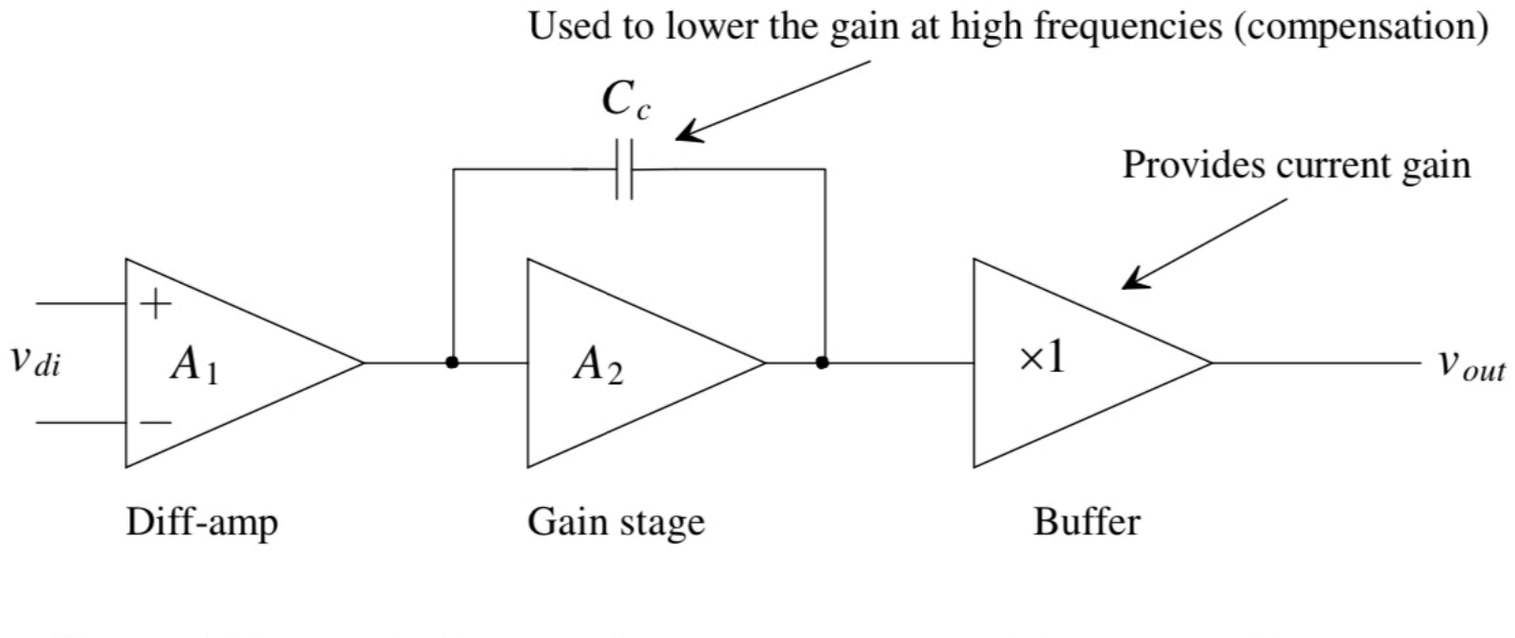


Figure 24.1 Block diagram of two-stage op-amp with output buffer.

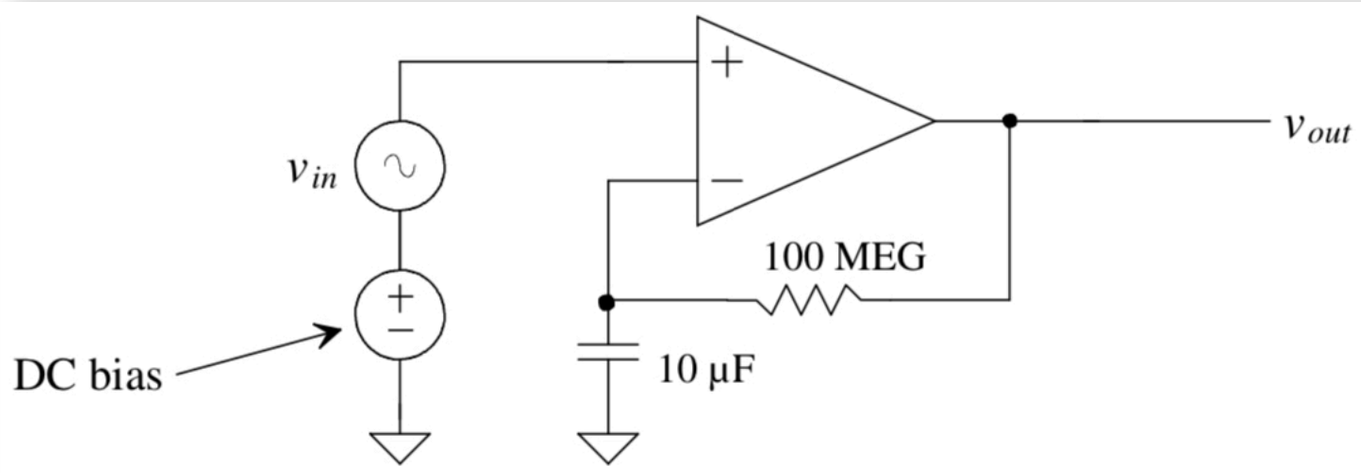


Figure 24.9 Circuit configuration used to simulate open-loop frequency response.

Simulation test- benches :

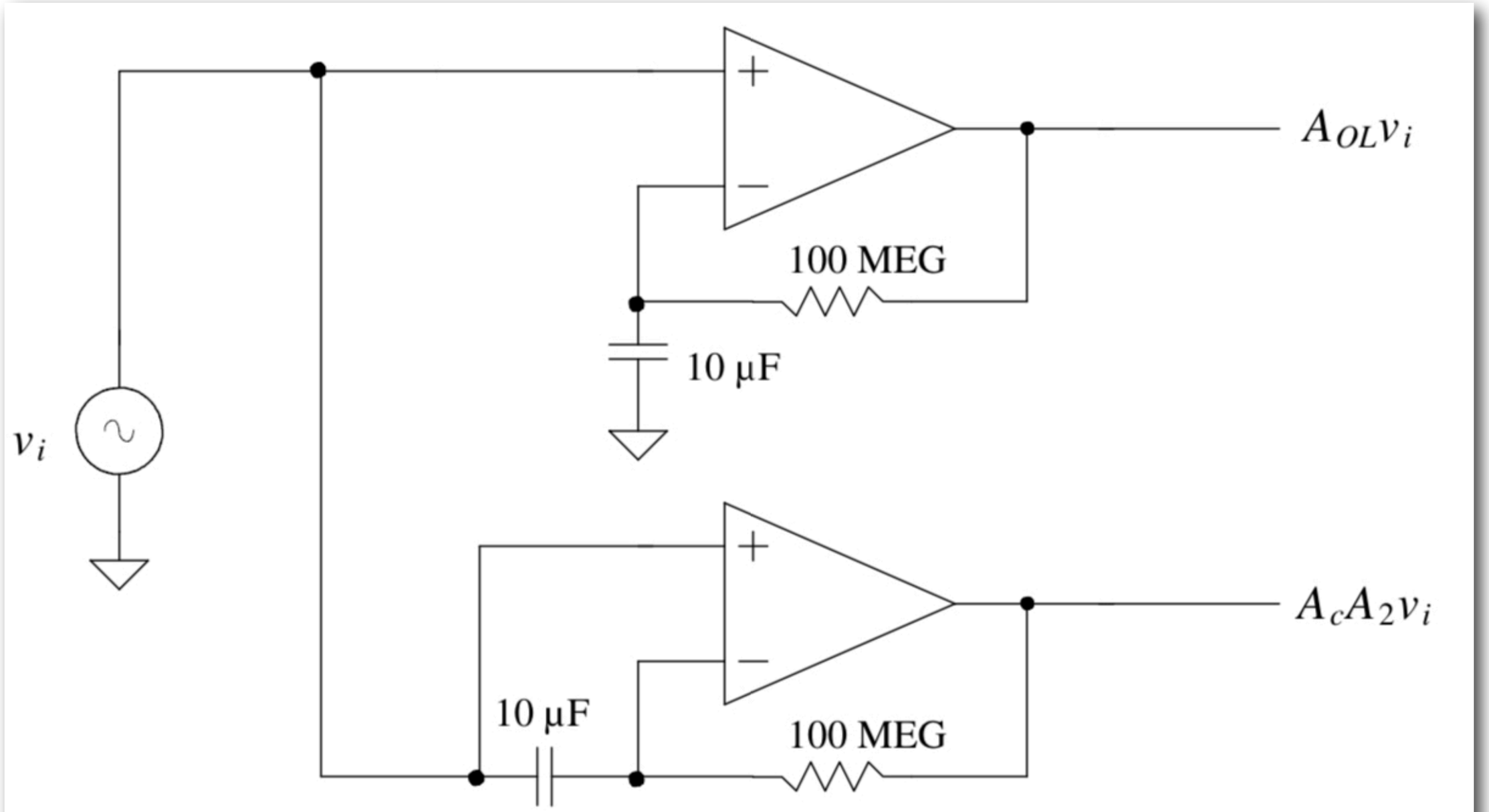


Figure 24.25 Circuit configuration used to simulate CMRR.

Simulation test-benches:

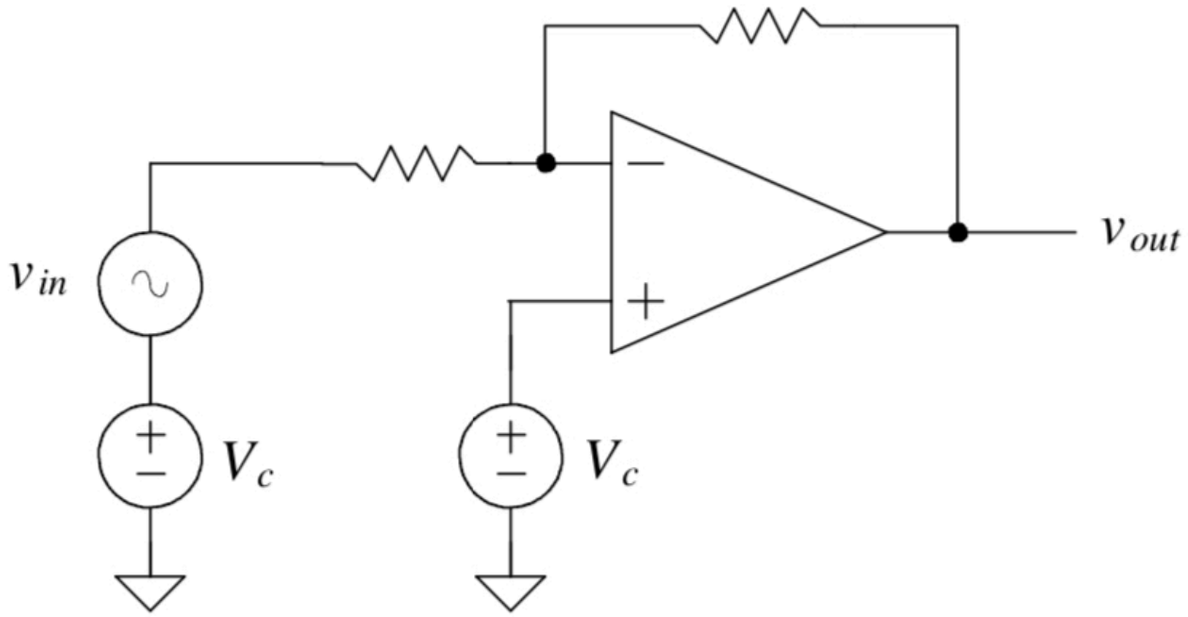


Figure 24.26 An inverting op-amp topology. The common mode voltage is held constant.

Simulation test-benches:

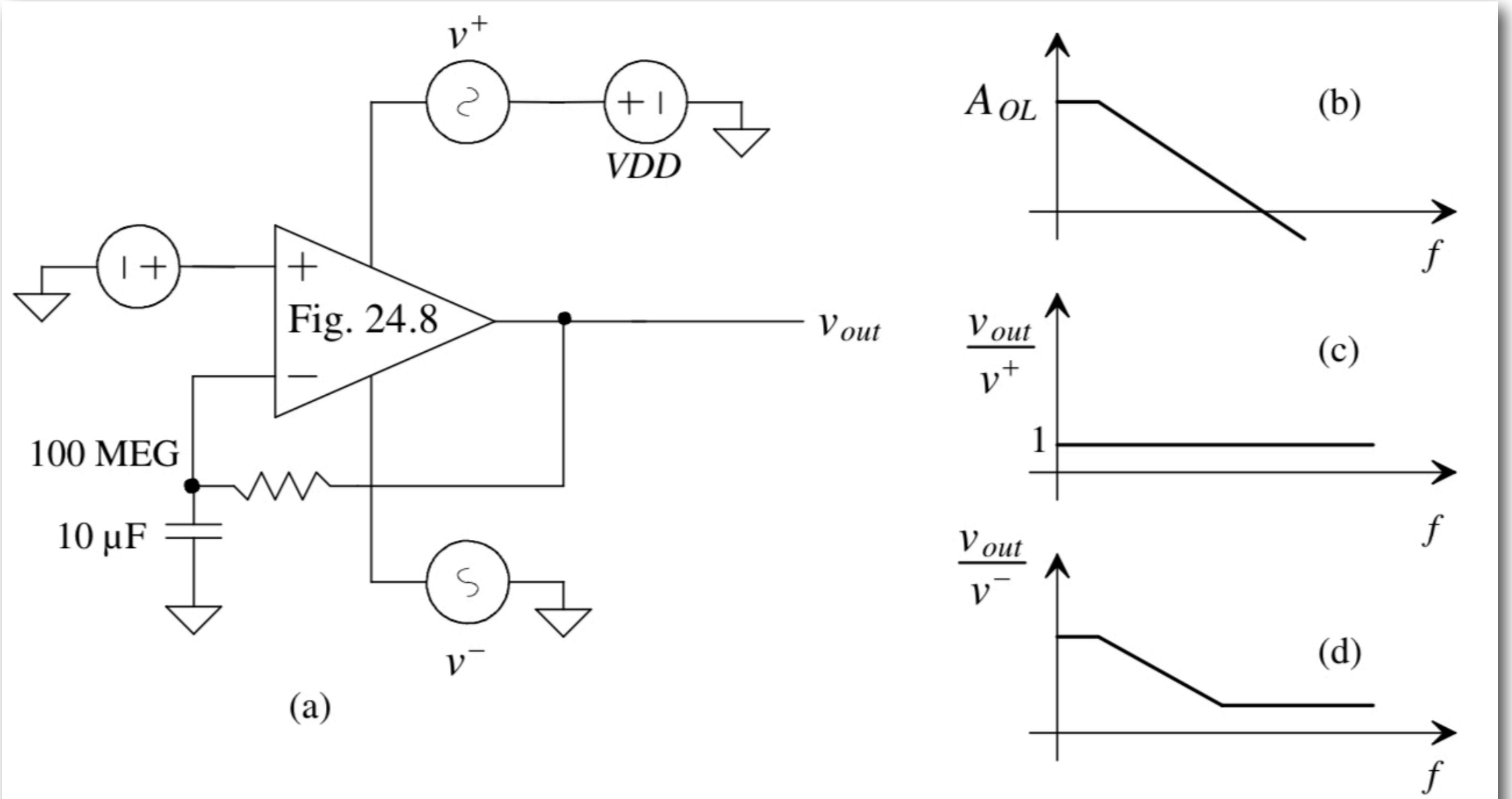


Figure 24.27 (a) Test setup to determine PSRR, (b) open-loop gain, (c) gain from AC signal on VDD to output, and (d) gain from AC signal on ground to output.

Thank You!